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Assignment - 1

1. Explain linux directory structure and file types in detail.

⇒ Linux Directory Structure :-
In, linux everything is treated as a file even if it is normal file, a directory, or even a device such as printer or keyboard. All the directories and files are stored under one root directory which is represented by forward slash /. The linux directory layout follows the filesystem Hierarchy Standard (FHS). This standard defines how directories are organized and what type of file should be stored in each.

* Types of files in the linux system.

Linux categorizes files into different types based on their purpose, such as general data storage.

1. General Files :-
It is also called ordinary files. it may be an image, video, program or simple text file. These types of files can be in ASCII or Binary format. It is most commanly used file in the linux system.

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2. Directory Files :-

These files are a warehouse for other files types. It may be a directory file within a directory. (subdirectory)

3. Device Files :-

In a windows like oprating system, devices like CD-ROM and hard drives are presented as drive letters like F: G: H, whereas in linux system, devices are represented as files. As for eg, /dev/sda1, /dev/sda2 and so on.

* Linux File Hirerarchy Structure.

The Linux File Hirarchy structure or the Filesystem Hierarchy Standard (FHS) defines the directory structure and directory cotentents in unix-like operating system. it is maintained by the linux Foundation.

1. / (Root):

At the top of every linux file system is the root directory represented by a forward slash /

its the base point, and no directory exists above it. if you look at the file systems graphically, all other directorieys branching from root directory

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- Every single file and directory start from the root directory.

- The only root user has the right to write under this directory.

- /root is the root user's home directory which is not the same as /

- Only the root user has permission to modify contents inside this directory

2. /bin :

The bin directory contains essential commands and binaries needed by all users, including cp, ls, ssh, and kill. These commands are universaley available across user types.

- Contains binary executables.

- Common linux commands you need to use in single-user modes are located under this directory.

- Commands used by all the users of the system are located here.
eg :- ps, ls, ping, grep, cp.

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3. ./boot :
This directory stores all files required for booting the system. It includes the GRUB boot-loader configuration and essential kernel files that are loaded during startup.

- kernel initrd, vmlinux, grub files are located under /boot.

- Eg:- initrd.img-2.6.32-34-generic, vmlinuz-2.6.32-24-generic.

4. ./dev :
Device files in linux are stored in the /dev directory. These are special files that act as interface between hardware and software. Devices files are block device & character devices.

- These includes terminal devices, usb, or any devices attached to system.

5. ./etc :
Short for "Editable Text Configuration" /etc contains configuration files for system application, users, service, and tools or it contains the host-specific system-wide configuration files.

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2. Explain Command to change file permission and ownership.

⇒ Change File Permission :-

Syntax :- chmod [options] mode file.

Eg :- Numeric Mode :-
chmod 755 file.txt.

This sets :- owner : read, write, execute(7
Group :- read, execute (5)
Others : read, execute (5)

Eg Symbolic Mode :-
chmod u+x file.sh.

Adds execute (+x) permission to user / owner (u)

⇒ Change File Ownership :-

Syntax :- chown [options] owner[:group] file

Eg: chown sonil file.txt

(Changes owner of file.txt to sonil).

Eg chown sonil : users file.txt

(Changes owner to sonil and group to users.).

Assignment - II

1 Explain grep and sed command in detail.

⇒ grep stands for Global Regular Expression Print).

Syntax: grep[options]'pattern' filename

- pattern - The text or regular expression you want to find
- filename - The file(s) to search in
- options - Modify the behaviour of grep.

⇒ -i :- Ignore case when matching. Uppercase = lowercase.

Eg:- grep -i "navsari" Student.lst

⇒ -v :- Invert match : shows lines that don't match the pattern.

Eg: grep -v "Bharuch" Student lst.

⇒ -n : Show line numbers where matches occur.

⇒ -c :- Count how many lines match the pattern.

Eg:- grep -c "Ahemdabad" Student lst.

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=> -l :- Print only the filenames containing the pattern.

=> -e :- Search for multiple patterns.

Eg: grep -e "Surat" -e "Baroda" -e "Vapi" student.lst

=> -f :- Read search patterns from a file.

eg: grep -f pat.lst student.lst

* Sed stands for Stream Editor, Sed is powerful Unix Utility that processes and transforms text files line-by-line using patterns based commands.

Syntax :- sed [options] 'address command' filename.

address :- A line number, range (eg, 1,5), or pattern (/pattern/).

Command :- Opration such as print (p), delete (d), Substitute (s)

Filename :- file you want to edit.

=> -n :- Suppress automatic printing pattern space.

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-e :- Add multiple commands in a single sed execution

-f :- Read commands from file.

--version :- Display sed version

--help :- Display help message.

* Commands in Sed

=> p :- Print line.

=> d :- Delete line

=> q :- Quit after N lines

=> i\text :- Insert text before a line.

=> s/pat/rep/ :- Substitute

=> w file : Write a file.

=> /pattern/d :- Match pattern

=> $!p : print all the lines except the last one (with -n)

=> 10,20/-/:/ :- In line 10 to 20, replace the ~~las~~ first dash - with colon :

2. Explain input/output redirection and command substitution.

⇒ Shell metacharacters are special characters that have a specific meaning to the shell. They help in satisfying and controlling command behaviour, such as file selection, redirection, and combining commands.

* Input / output Redirection.

⇒ Redirection is used to send input/output of a command to/from a file instead of a screen.

(Output Redirection) :- Redirects output to file, overwriting the file.
Eg:- echo "Hello" > output.txt

(Append output) :- Redirects output and appends to the file without overwriting
Eg:- echo "world" >> output.txt

(Input Redirection): Takes input file instead of keyboard.
Eg: sort < names.txt

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* Command Substitution ($(...), `...`) :-
Used to capture the output of a command and use in it as an argument in another command.

Syntax :- 1: $(...) (Recommended).
today = $(date)
echo " Today is : $today "

Syntax :- 2 : `...` (Backticks).
echo " Today is : `date` "

Both versions store the output of date into a variable or inline expression.